GRADUATION PROJECT

Human Resources Dataset Analysis

Project Plan & Management Report

Project Type	Data Analysis Graduation Project
Team Size	5 Members
Main Tools	Power BI · Power Query · Excel · SQL/Python if needed
Project Start	16 October 2026
Submission Deadline	30 November 2026

Prepared by:

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Executive Summary

This project will analyze the Human Resources dataset to uncover practical insights into the workforce, including employee characteristics, compensation, promotions, performance, and attrition. The team will prepare and transform the data in Power Query, use Excel for exploration and validation, and build interactive dashboards in Power BI. SQL or Python may be used where they add value.

All five team members will contribute to data cleaning and preparation. Once the team approves a clean, reliable dataset, each member will develop one dashboard. Every dashboard will then undergo peer review before final revisions. The project will run from 16 October to 30 November 2026, with dedicated time for analysis, review, documentation, presentation preparation, and final quality checks.

The project will prioritize substance as well as presentation. Its purpose is to answer meaningful HR questions, support conclusions with evidence, and translate the findings into practical recommendations grounded in the available data.

Project Proposal

Overview and Background

HR departments manage information about employees, compensation, promotions, performance, and turnover. When analyzed effectively, this information can reveal workforce patterns and highlight areas that require attention. This project will examine the available HR data and communicate its findings through interactive Power BI dashboards.

Problem Statement

Raw HR data stored in tables or spreadsheets can be difficult to interpret and use. The team will determine how structured analysis and clear visualization can make the data more accessible, reveal meaningful patterns, and answer practical HR questions.

Business Objectives

- Understand the dataset and document its available fields.
- Identify data-quality problems and prepare a reliable dataset.
- Explore employee demographics and workforce characteristics where supported.
- Examine salary distribution, promotions, performance, and attrition when the required fields exist.

• Build five Power BI dashboards, one for each member.

• Validate calculations and findings before presenting them.

• Provide evidence-based insights and practical recommendations.

• Document the process and present the results professionally.

Scope

- Inspect the dataset and create a data dictionary.
- Assess data quality and clean the data collaboratively in Power Query.
- Conduct exploratory data analysis and define feasible business questions.
- Develop the data model and DAX measures where required.
- Build and peer-review five dashboards.
- Validate findings and prepare the final report and presentation.

Expected Outcomes and Business Value

The project will produce a cleaned dataset, a data dictionary, a cleaning log, an approved set of business questions, documented analytical findings, five peer-reviewed dashboards, evidence-based recommendations, a final report, and a presentation. Together, these deliverables will make the HR information easier to explore and help decision-makers identify patterns that warrant further investigation.

Project Scope and Business Questions

Area	Business question	Required data or condition
Workforce overview	How many employees are represented?	Reliable employee identifier or agreed counting method
Demographics	What employee groups are represented?	Relevant demographic fields
Workforce distribution	How are employees distributed across departments, roles, or locations?	Relevant group fields
Salary	What is the salary distribution? Do values differ between groups?	Salary and a suitable grouping field
Promotions	How many employees have promotion records, and what patterns appear?	Promotion status, dates, or event history
Performance	How are available performance ratings distributed?	Performance rating or evaluation data
Attrition	What proportion of records is marked as attrition?	Reliable attrition or employment-status field
Retention	Which groups show possible retention concerns?	Suitable employment-history data

Relationships	Are there patterns between salary, promotion, performance, and attrition?	Relevant fields for the variables compared

Detailed Task Assignment for All 5 Members

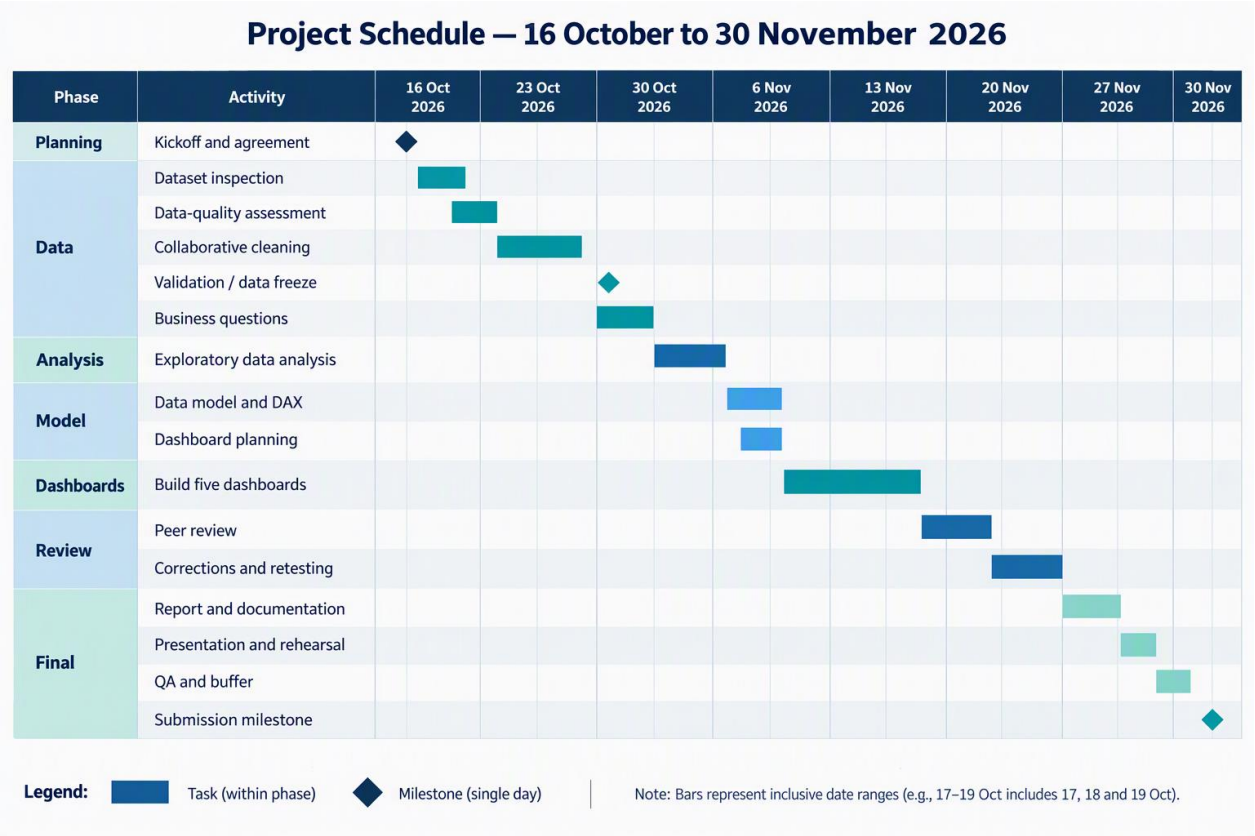
Team Roles

Member	Main responsibility	Provisional dashboard topic
Hebatallah Ashraf	Project coordinator, schedule, task tracking, final submission	Workforce overview and demographics
Hasnaa sobhy	Dataset inventory, cleaning-log coordination, transformation documentation	Salary and compensation
Esraa Mahmoud	Data-quality assessment, EDA coordination, analytical validation	Promotions and performance
Nehad Adel	Data model, DAX support, technical review	Attrition and retention
Rawan Abdelrahman	Report and presentation coordination, documentation and file organization	HR summary and key findings

Peer Review Assignments

Dashboard owner	Reviewer	Deadline
Hebatallah Ashraf	Rawan Abdelrahman	17 Nov
Hasnaa sobhy	Hebatallah Ashraf	17 Nov
Nehad Adel	Hasnaa sobhy	17 Nov
Esraa Mahmoud	Nehad Adel	17 Nov
Rawan Abdelrahman	Esraa Mahmoud	17 Nov

Gantt Chart



Milestones and Deliverables

Milestone	Deadline	Evidence
Project plan approved	16 Oct	Roles, dates, communication rules
Dataset understood	19 Oct	Inventory and draft dictionary
Quality assessment complete	21 Oct	Issue list and cleaning proposals
Cleaning complete	26 Oct	Cleaned dataset and log
Clean data approved	27 Oct	Validation checks and saved version
Business questions approved	29 Oct	Feasible question list

EDA complete	2 Nov	Summary tables and initial findings
Model and dashboard plans approved	5 Nov	Model, measures, five outlines
Five dashboards complete	13 Nov	First-version dashboards
Peer reviews complete	17 Nov	Five review checklists
Corrections complete	21 Nov	Updated dashboards and issue log
Report finalized	25 Nov	Complete report and documentation
Presentation approved	27 Nov	Slides and rehearsal
Final QA passed	29 Nov	Completed checklist and final files
Project submitted	30 Nov	Submission evidence

Risk Assessment and Mitigation Register

Risk	Prevention	Contingency
Missing or poor-quality data	Profile fields early; log missingness	Reduce scope to supported questions and state limitations
Cleaning delays	Agree rules first; review progress daily	Prioritize analysis-critical issues and use buffer
Wrong types or inconsistent values	Check types, categories, dates, and numbers	Reprocess fields and rerun validation
Unclear business questions	Specify fields and expected output for each question	Rewrite/remove unsupported questions by 29 Oct
Incorrect calculations or conclusions	Document definitions and independent checks	Correct measures, revise conclusions, retest
Power BI/DAX issues	Model early; validate measures independently	Simplify model or calculation and document limits
Unequal participation	Visible owners, deadlines, weekly tracking	Reassign specific support tasks by agreement
Missed deadlines	Track upcoming deadlines and blockers	Use buffer and reduce low-priority features
Conflicting file versions	Single source of truth and version naming	Restore approved version and document changes
Late report or presentation	Draft documentation throughout project	Focus on required content and rehearse essentials

Inconsistent dashboard definitions	Use shared metric definitions and model	Reconcile definitions and repeat review
Sensitive data exposed	Restrict access and avoid public personal identifiers	Remove exposed file, restrict access, replace approved copy

Project Management KPIs

- Tasks completed on time (%)
- Milestones completed on time (%)
- Data-quality checks passed (%)
- Unresolved high-priority issues (#)
- Business questions answered with evidence (%)
- Independently validated insights (#)
- Actionable, evidence-based recommendations (#)
- Dashboards peer-reviewed (%)
- Dashboard calculations verified (%)
- Dashboard features functioning correctly (%)
- Required documentation approved (%)
- Assigned team tasks completed (%)
- Peer reviews completed on time (%)
- Submission requirements quality-checked (%)

2. Literature Review

1. Introduction

Human Resources Analytics (HR Analytics) involves analyzing employee-related data to understand workforce characteristics, identify patterns, and support better human resource management decisions. Organizations can use HR analytics to examine employee attrition, satisfaction, performance, demographics, compensation, and working conditions.

Power BI is a useful tool for this purpose because it enables analysts to transform raw data into interactive dashboards, calculate key performance indicators (KPIs), and present findings through charts, tables, and filters.

Before developing our project, we reviewed several existing HR Analytics dashboard projects to understand how other analysts organize HR data, select relevant KPIs, and communicate their findings. This review helped us identify useful design practices, potential limitations, and opportunities to improve our own dashboard.

The reviewed projects are examples of previously published HR Analytics work. They are used as references for dashboard design and analytical approaches, rather than as evidence that a particular company adopted or implemented them.

2. Review of Previous Projects

Project 1: HR Analytics Dashboard — Aakash Singh Dahiya

Reference: https://www.aakashsinghdahiya.com/projects/hr-analytics-dashboard

This project presents an HR Analytics dashboard developed using Power BI. It focuses on understanding employee attrition and exploring workforce characteristics through visual analysis.

Main areas of interest:

- Employee attrition and workforce distribution.
- Employee demographics and job-related characteristics.
- Factors and categories that can be explored in relation to employee turnover.
- Presenting HR indicators through an interactive visual dashboard.

Strengths:

- Provides a visual approach to understanding employee attrition.
- Makes workforce patterns easier to explore than examining raw data alone.
- Demonstrates how HR metrics can be organized into a dashboard for analysis.

Lessons learned for our project:

This project demonstrates the importance of including attrition indicators and relevant employee characteristics in an HR dashboard. We can apply this idea by presenting the attrition rate, employee counts, and comparisons across relevant employee groups. We should also explain the findings carefully and avoid presenting associations as proven causes.

Project 2: Atlas Labs HR Analytics Dashboard — Shayan Razavi

Reference: https://shayanrazavi.com/project5/

This project presents an HR Analytics dashboard focused on examining employee information and workforce-related indicators. It provides an example of how employee data can be organized into visual reports to support exploration and interpretation.

Main areas of interest:

- Employee-related metrics and workforce patterns.
- Exploring HR information through visualizations.
- Organizing analytical results into a dashboard format.

Strengths:

- Demonstrates the value of visual storytelling in HR analysis.
- Helps users explore workforce information through a structured presentation.
- Provides a reference for organizing HR indicators into a more accessible analytical interface.

Lessons learned for our project:

This project highlights the importance of organizing the dashboard around clear analytical questions. In our project, we will use separate dashboard pages for workforce overview, attrition, employee satisfaction and performance, compensation and work-life balance, and employee profiles. This structure will help users move from general workforce information to more specific findings.

Project 3: HR Analytics Power BI Dashboard — maw-khan

Reference: https://github.com/maw-khan/HR-Analytics-PowerBI-Dashboard

This project is a publicly available Power BI HR Analytics dashboard project. Its repository provides an additional reference for examining how HR information can be prepared and presented through a business intelligence solution.

Main areas of interest:

- HR data visualization using Power BI.
- Presenting workforce indicators in a dashboard.
- Applying business intelligence techniques to employee-related information.

Strengths:

- Provides a practical example of a Power BI implementation.
- Offers a reference for dashboard organization and the presentation of HR indicators.
- Can help analysts understand how a dashboard project is structured and delivered.

Lessons learned for our project:

This project reinforces the importance of combining dashboard design with accurate calculations and meaningful analysis. Therefore, our team will validate the measures, check relationships between tables, and ensure that every visualization serves a clear analytical purpose.

Project 4: HR Analytics Power BI — Ediomo Nyong

Reference: https://github.com/ediomonyong/HR_Analytics_PowerBI

This project is another publicly available example of HR Analytics implemented with Power BI. It provides a reference for exploring the application of data visualization and business intelligence techniques to HR-related information.

Main areas of interest:

- Analyzing employee data through Power BI.
- Presenting HR information in a visual format.
- Using dashboard-based analysis to explore workforce indicators.

Strengths:

- Demonstrates a practical application of Power BI to HR data.
- Provides another example of how employee information can be communicated visually.

- Offers an additional reference for reviewing dashboard implementation approaches.

Lessons learned for our project:

This project emphasizes the need to balance implementation, analytical relevance, and communication. We will aim to make our dashboard understandable to users who may not have a technical background, while retaining enough detail to support meaningful HR analysis.

Comparison Between Existing Projects and Our Project

The reviewed projects demonstrate different examples of applying Power BI to HR Analytics. Although their precise datasets, measures, and layouts may differ, they provide useful references for our project in terms of visualization, KPI presentation, and the organization of HR information.

Our project uses an HR dataset containing employee-related attributes, including demographic, job-related, satisfaction, performance, and compensation information. Our objective is to use descriptive analysis to explore workforce patterns and present the results in an interactive Power BI dashboard.

Comparison aspect	**Existing projects**	**Our project**
Main objective	Analyze and visualize HR-related information	Explore employee data and support HR decision-making through descriptive analysis
Data preparation	Depends on each project's dataset and workflow	We will clean, transform, and validate our assigned dataset before analysis
Attrition analysis	Included where relevant to the individual project	We will examine attrition levels and compare them across relevant employee groups
Employee satisfaction	Depends on the variables and measures selected	We will explore available satisfaction indicators and their relationship with other employee attributes
Performance analysis	Depends on each project's analytical scope	We will analyze available performance ratings and compare them with relevant employee characteristics
Compensation and work-life balance	Depends on the available data and dashboard scope	We will explore compensation-related variables, overtime, and work-life balance where supported by the dataset
Dashboard structure	Varies by project	We plan to organize our dashboard into dedicated pages based on our analytical questions

Interactivity	Depends on the individual implementation	We will use appropriate slicers, filters, and interactive visuals
Analytical approach	Varies according to the project's objectives	Descriptive analysis using Power BI, DAX measures, and visualizations

Research Findings and Lessons Learned

The literature review led to several lessons that will guide our project:

1. **Define clear analytical questions. Each dashboard page should answer specific questions rather than displaying charts without a clear purpose.**
2. **Prioritize data quality. Missing values, duplicates, inconsistent categories, and incorrect data types can affect the accuracy of the analysis.**
3. **Use meaningful KPIs. Measures such as employee count and attrition rate should be clearly defined and validated.**
4. **Organize the dashboard logically. A general workforce overview can introduce the analysis, followed by more detailed pages on attrition, satisfaction, performance, compensation, and work-life balance.**
5. **Use interactivity appropriately. Slicers and filters should help users explore the data without making the dashboard unnecessarily complicated.**
6. **Interpret findings carefully. A relationship between two variables does not necessarily establish a causal relationship.**
7. **Validate both calculations and visuals. Accurate DAX measures, appropriate relationships, and consistent filter behavior are essential for trustworthy results.**
8. **Connect findings to HR decisions. The final report should explain what the observed patterns may indicate and what additional investigation may be useful.**

2. Feedback & Evaluation

Project Evaluation Approach:

As part of our internal project review, we will evaluate the quality of our HR Analytics solution throughout its development. The evaluation will focus on whether the dataset has been prepared correctly, whether the analysis answers the intended questions, whether the dashboard is usable, and whether the findings are supported by the available data.

Our evaluation will be carried out by the project team through review, testing, and discussion. It represents our own assessment process and should not be presented as feedback already received from the lecturer or an external evaluator.

Evaluation Criteria and Validation Methods:

We will evaluate the project using the following internal checks:

Evaluation area	How we will evaluate it

Data preparation	Inspect data types, missing values, duplicates, and inconsistent categories; document cleaning decisions
Data model	Review table relationships and confirm that filters behave as expected
DAX measures	Recalculate selected KPIs independently and compare them with Power BI results
Analytical relevance	Check that each visualization answers a defined HR question
Visual design	Review titles, labels, layout, readability, and consistency
Interactivity	Test slicers, filters, and cross-filtering across relevant visuals
Interpretation	Confirm that written findings are supported by the data and avoid unsupported causal claims
Documentation	Ensure that data preparation, measures, analytical findings, limitations, and decisions are documented
Presentation	Check that the team can explain the dashboard, its KPIs, and its main findings clearly

3. Suggested Improvements

Based on the literature review and our internal evaluation plan, we identified several improvements that can strengthen the quality and usefulness of our project

1. Improve Data Quality and Preparation

Before analysis, we will inspect the dataset for missing values, duplicates, inconsistent categories, and incorrect data types. We will document the cleaning decisions and verify that any changes preserve the meaning of the original data.

Where appropriate, we will also check whether the dataset contains fields that should be treated as categories rather than numerical measurements. This will help ensure that the data is modeled and visualized correctly.

2. Establish Clear KPI Definitions

We will define each KPI before using it in the dashboard. For example, the attrition rate must use a clearly specified numerator and denominator, and employee counts must respond correctly to the selected filters.

We will use DAX measures where appropriate and validate the results against the source data. This will help reduce inconsistencies between dashboard pages and improve confidence in the results.

3. Organize Dashboard Pages Around Analytical Questions

We plan to divide the dashboard into pages with distinct purposes:

- **HR Overview: Provide an overall view of the workforce and key HR indicators.**
- **Employee Attrition Analysis: Explore attrition patterns across relevant employee groups.**
- **Employee Satisfaction & Performance: Examine available satisfaction and performance measures.**
- **Compensation & Work-Life Analysis: Explore compensation-related variables, overtime, and work-life balance where supported by the dataset.**
- **Employee Profile & Recommendations: Present relevant employee characteristics and summarize findings that may support further HR investigation.**

4. Enhance Visual Clarity and Interactivity

We will use clear titles, consistent formatting, readable labels, and appropriate chart types. Slicers will be selected based on their usefulness to the intended analysis, rather than added simply for decoration.

We will also avoid overcrowding the pages with too many charts. Each visual should communicate a meaningful finding or help users answer a specific question.

5. Strengthen Testing and Validation

Before final submission, we will test the dashboard using different filter selections and check whether the relevant KPIs update correctly. We will verify the behavior of relationships, measures, and visuals and correct any discrepancies.

We will also review the final report against the dashboard to ensure that the written findings match the displayed results.

6. Improve Interpretation and Recommendations

We will explain the meaning of the observed patterns rather than only reporting numerical values. For example, if attrition differs between groups, the report will describe the difference and suggest areas for further investigation without claiming that the group characteristic necessarily caused the difference.

Recommendations will be based on the actual results of the completed analysis. We will distinguish between findings directly supported by the dataset and possible actions that require additional investigation.

7. Improve Team Collaboration and Documentation

The team will maintain consistent definitions for measures, document important data-cleaning decisions, and coordinate edits to the Power BI project. Each member will contribute to reviewing calculations and dashboard outputs rather than limiting their involvement to a single isolated task.

This approach will support consistency across the dashboard, documentation, and final presentation.

4. Final Grading Criteria

To maintain quality throughout the project, our team will use an internal assessment framework to review the completeness and quality of the deliverables. These criteria will help us identify weaknesses before submission and ensure that the project meets its intended objectives.

Proposed Assessment Criteria :

Assessment criterion	What we will evaluate
Documentation and Literature Review	Clear project objectives, relevant review of previous projects, appropriate references, documented methodology, and clear explanations
Data Cleaning and Preprocessing	Correct data types, appropriate handling of missing values and duplicates, consistent categories, and documented preparation decisions
Data Modeling and DAX	Appropriate relationships, clearly defined measures, correct calculations, and validation of key KPIs
Analytical Implementation	Relevant analytical questions, appropriate use of the available variables, and meaningful interpretation of results
Dashboard Design and Functionality	Logical page organization, readable visuals, useful slicers, consistent formatting, and correct filter behavior
Testing and Validation	Checks of calculations, relationships, filters, and consistency between the dashboard and the report
Findings and Recommendations	Conclusions supported by the data, clear explanations of limitations, and practical recommendations where justified
Final Report and Presentation	Organized reporting, clear communication of the results, appropriate visual support, and the ability to explain the analytical process

References

1. Dahiya, A. S. *HR Analytics Dashboard*. https://www.aakashsinghdahiya.com/projects/hr-analytics-dashboard
2. Razavi, S. *Atlas Labs HR Analytics Dashboard*. https://shayanrazavi.com/project5/
3. maw-khan. *HR Analytics Power BI Dashboard*. GitHub. https://github.com/maw-khan/HR-Analytics-PowerBI-Dashboard

4. Nyong, E. *HR Analytics Power BI*. GitHub. https://github.com/ediomonyong/HR_Analytics_PowerBI

3. Requirements Gathering

1. Stakeholder Analysis

Stakeholder analysis identifies the people or groups who may benefit from the project and defines their main needs. Our HR Analytics Dashboard uses Power BI to analyze employee data and present insights about workforce characteristics, attrition, satisfaction, performance, compensation, and work-life balance.

The main stakeholders and their needs are summarized below.

Stakeholder	Needs and Expectations
HR Managers	Monitor employee attrition, satisfaction, performance, and workforce indicators to support HR decisions.
HR Analysts	Explore employee data, compare different groups, and identify patterns using filters and visualizations.
Senior Management	Review key HR indicators and summarized findings to support management discussions.
Project Team	Clean the dataset, develop DAX measures, build the dashboard, validate results, and prepare documentation.
Academic Supervisor/Lecturer	Review the project methodology, dashboard functionality, analytical findings, and final deliverables.

The dashboard is primarily designed to support HR analysis by presenting clear and interactive information. Its findings will help users identify patterns that may require further investigation.

2. User Stories & Use Cases

1. User Stories

User stories describe what users need from the dashboard and why these features are useful.

- **User Story 1 – HR Overview:** As an HR manager, I want to view key workforce indicators so that I can understand the overall employee situation.
- **User Story 2 – Attrition Analysis:** As an HR analyst, I want to compare attrition across employee groups so that I can identify patterns and areas requiring further investigation.
- **User Story 3 – Satisfaction and Performance:** As an HR manager, I want to explore employee satisfaction and performance indicators so that I can understand differences between employee groups.

- **User Story 4 – Interactive Filtering:** As a dashboard user, I want to filter results by relevant employee attributes so that I can focus on specific groups and compare their results.

2. Use Cases

Use Case 1: View HR Overview

- **Actor:** HR Manager.
- **Goal:** Understand the overall workforce situation.
- **Main steps:**
    1. The user opens the HR Overview page.
    2. The dashboard displays the available workforce KPIs and charts.
    3. The user reviews the indicators and applies filters if needed.
    4. The dashboard updates the relevant results.
- **Expected result:** The user obtains a clear overview of the workforce based on the available data.

Use Case 2: Analyze Employee Attrition

- **Actor:** HR Analyst.
- **Goal:** Explore attrition patterns across employee groups.
- **Main steps:**
    1. The analyst opens the Employee Attrition Analysis page.
    2. The dashboard displays attrition metrics and related visualizations.
    3. The analyst selects a relevant filter, such as department or overtime status, if available.
    4. The analyst compares the results and records the main observations.
- **Expected result:** The analyst identifies descriptive patterns that may support further HR investigation.

3. Functional Requirements

Requirement	Description
Data Import	Load the provided HR dataset into the appropriate analysis environment.
Data Cleaning	Identify and handle relevant missing values, duplicates, inconsistent categories, and incorrect data types.

Data Modeling	Organize the data and establish appropriate table relationships where required.
KPI Calculation	Create and validate relevant DAX measures, such as employee count and attrition rate.
HR Overview	Display the main workforce indicators in an overview page.
Attrition Analysis	Visualize attrition across relevant employee groups and available attributes.
Satisfaction and Performance	Present visualizations of available employee satisfaction and performance indicators.
Compensation and Work-Life Analysis	Explore available compensation, overtime, and work-life balance variables.
Interactive Filters	Allow users to filter relevant dashboard results using slicers and filters.
Reporting and Documentation	Present the main findings and document the data preparation, analysis, and project results.

The implementation of each requirement will depend on the fields available in the dataset.

The project focuses on descriptive analysis and visualization.

4. Non-functional Requirements

Non-functional requirements define the quality standards that the dashboard should meet.

Category	**Requirement**
Performance	The dashboard should respond within a reasonable time when users navigate between pages or apply filters.
Security and Privacy	Employee data must be handled responsibly, and unnecessary personal information must not be exposed.
Usability	The dashboard should use clear titles, readable labels, and appropriate visualizations.
Reliability	KPIs and visualizations should display consistent results for the same data and filter conditions.
Accuracy	Calculations and findings must be checked against the source data and must not include unsupported conclusions.
Maintainability	Important DAX measures, data transformations, and relationships should be documented clearly.

Compatibility	The Power BI project and supporting files should open correctly in the team's agreed working environment.

These requirements will be checked through data validation, testing of DAX measures, review of dashboard interactions, and inspection of the final deliverables.